جامعة خليفة
Khalifa University

MYOSA Event - IEEE Sensors Council

KINESIS: An AI-Driven Wearable Rehabilitation Monitoring System Using the MYOSA Platform for Predictive Recovery Analytics

IEEE MYOSA Event 5.0 | IEEE Biosensors 2026

A. Expression of Interest

We express our strong interest in participating in the IEEE International MYOSA Event 5.0. Our team proposes KINESIS (Kinetic Intelligence for Neuromusculoskeletal Evaluation, Synthesis, and Insight Systems), an AI-driven wearable rehabilitation monitoring platform built on the MYOSA Mini Kit. The system uses the kit's MPU6050 as the primary motion sensor alongside the APDS9960, BMP180, and SSD1306 OLED to capture and classify patient movement quality during rehabilitation exercises. A companion smartphone application performs real-time AI inference using TensorFlow Lite, providing clinicians and patients with objective recovery metrics, movement quality scores, and predictive alerts. The project addresses a critical global healthcare challenge: musculoskeletal disorders affect 1.71 billion people worldwide [1], yet rehabilitation monitoring remains subjective and discontinuous. KINESIS transforms the MYOSA platform into a clinically meaningful sensor system, demonstrating its potential for healthcare IoT. Our team of four electrical engineering students at Khalifa University combines expertise in embedded systems, IoT, AI/ML, and healthcare technology.

B. Team Member Details

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D. Ideation Details

D.1 Problem Statement

Musculoskeletal (MSK) disorders affect approximately 1.71 billion people globally and are the leading contributor to disability across 160 countries [1]. The Global Burden of Disease Study 2019 established that MSK conditions account for two-thirds of all prevalent cases requiring rehabilitation, with low back pain alone affecting 568 million people [2]. The economic burden is severe: MSK disorders were responsible for a global value of lost welfare of $2,099.84 billion in 2021, representing 1.41% of global GDP [3]. By 2035, the number of MSK cases is projected to rise to 2.161 billion [4].

Despite this scale, rehabilitation monitoring relies on periodic in-clinic assessments conducted 2-3 times per week, each lasting 30-60 minutes. Between sessions, recovery progress is unobserved, creating blind spots where regressions, compensatory movements, or early complications go undetected. Clinical assessments of movement quality remain subjective and exhibit significant inter-rater variability. A 2026 JMIR systematic review confirmed that while IMU-based sensors were used in 67% of clinical rehabilitation studies and machine learning was frequently applied, these technologies have not been integrated into routine clinical workflows [5]. Furthermore, biomechanical assessment and diagnostic data (imaging, lab results) are analyzed separately, preventing the multi-modal correlations that research has shown to dramatically improve predictive accuracy: combining clinical and sensor data achieves a 0.94 correlation with outcomes versus 0.79 for clinical data alone [6].

D.2 Proposed Solution

KINESIS is an AI-driven rehabilitation intelligence platform that fuses wearable IMU-based motion capture with clinical data analysis to deliver continuous, objective, and predictive recovery monitoring. The system provides four integrated capabilities: (1) continuous biomechanical monitoring via body-worn IMU sensors capturing ROM, angular velocity, jerk, and symmetry metrics; (2) multi-modal clinical data fusion integrating sensor streams with imaging, lab results, and clinical records; (3) predictive recovery analytics generating individualized trajectory forecasts, complication risk scores, and milestone probabilities; and (4) explainable clinical reporting producing natural-language narratives that translate AI outputs into actionable clinical insights. The system evolves through three phases: Phase 1 uses the MYOSA kit with phone-based TFLite inference for proof-of-concept; Phase 2 upgrades to clinical-grade sensors (BNO085) with a Raspberry Pi 5 + AI HAT 2+ (Hailo-8L NPU, 26 TOPS) for on-device edge AI; Phase 3 delivers custom PCB sensor nodes and a cloud-based multi-modal fusion engine with hospital EHR/PACS integration.

D.3 Uniqueness of the Solution

KINESIS addresses a structured gap that no existing solution fills. Consumer digital MSK platforms (Sword Health at $4B valuation, Hinge Health with $574M revenue [7]) target employer-sponsored pain management but do not integrate clinical diagnostic data or provide predictive analytics for clinicians. Clinical motion capture systems (VICON, Xsens/Movella) provide research-grade biomechanics but cost $50K-$500K and operate in isolation from clinical data [8]. KINESIS uniquely combines: (a) multi-modal data fusion bridging biomechanical and clinical domains; (b) predictive rather than descriptive analytics, with research showing wearable sensors can predict fracture nonunion with 82% accuracy at 6 weeks post-surgery [9]; (c) explainable AI outputs aligned with clinical reasoning; and (d) privacy-preserving federated learning for multi-institutional model improvement. No existing commercial or research system jointly provides these four capabilities.

D.4 How the MYOSA Board Will Be Used

The MYOSA motherboard (ESP32) serves as the central processor for each wearable sensor node. Core 0 runs the Madgwick sensor fusion algorithm converting raw MPU6050 data into quaternion-based orientation at 50-100 Hz; Core 1 manages BLE streaming to the companion app. The I2C bus connects all kit sensors via plug-and-play stacking. Additional MPU6050 boards cascade using I2C address selection (AD0 pin) or TCA9548A multiplexer for multi-joint monitoring. The open-source firmware is customized via Arduino IDE to implement sensor fusion, feature preprocessing, BLE GATT services, and OLED rendering. Built-in Wi-Fi enables optional direct cloud upload.

D.5 Sensors and Actuators

MPU6050 (Accel+Gyro): Primary motion sensor for 6-axis joint kinematics, ROM, jerk, and symmetry. 2-3 units on limb segments.

APDS9960 (Gesture/Proximity/Light): Contactless session trigger (proximity) and exercise navigation (gestures) for clinical hygiene. Light sensor adapts OLED brightness.

BMP180 (Barometric Pressure): Activity context via altitude changes (~0.12 hPa/m) for sit-to-stand detection, a key functional mobility metric fused with IMU data.

SSD1306 OLED: Real-time display of exercise name, rep count, ROM degrees, quality score (0-100), and green/yellow/red status.

ESP32 Motherboard: Dual-core sensor fusion + BLE/Wi-Fi hub.

Additional: 1-2 extra MPU6050 breakouts (~$4 each), Li-Po 3.7V + TP4056 charger, 3D-printed enclosures with straps. No actuators used.

D.6 Additional Circuit and Software

Circuit: Extra MPU6050 via I2C (AD0 or TCA9548A multiplexer); Li-Po battery + TP4056; 3D-printed enclosures with anatomical strap mounts.

Software: Arduino firmware (Madgwick filter, multi-sensor I2C, BLE GATT, OLED driver). Android app (Kotlin): BLE reception, sliding-window feature extraction (ROM, jerk, symmetry, barometric sit-to-stand), TFLite 1D CNN classifier + ROM tracker, real-time dashboard, PDF report generation. Optional Firebase cloud backend.

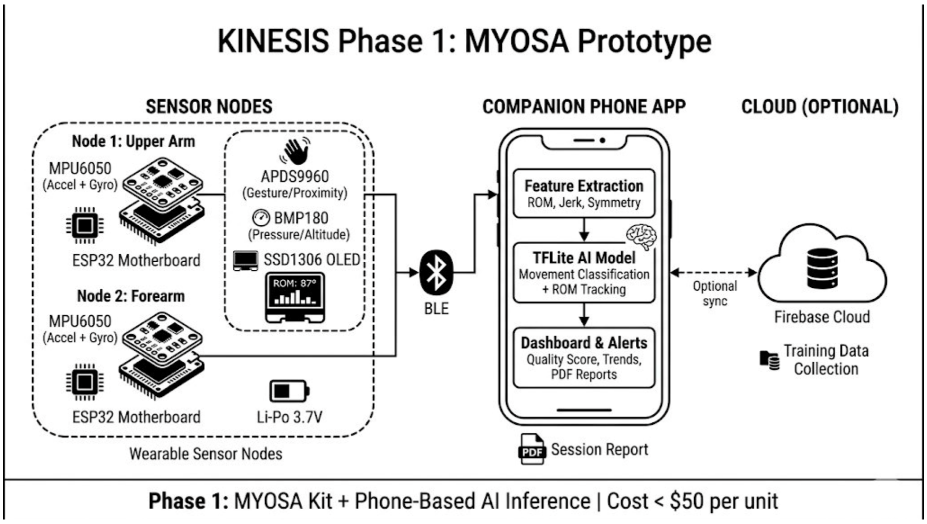

Fig. 1. Phase 1: MYOSA sensor nodes stream via BLE to phone-based TFLite AI inference.

D.7 Significance of the Application

MSK disorders are the leading cause of global disability, affecting 1 in 2 adults and costing $661 billion annually [10]. Current rehabilitation assessment is periodic, subjective, and siloed. KINESIS provides continuous, objective, AI-driven monitoring using affordable wearables, enabling predictive rehabilitation that can reduce complication rates, shorten recovery timelines, and lower costs. The UAE healthcare sector (AED 70B expenditure) with world-class hospitals presents an ideal deployment environment [11].

D.8 Exploitation of the MYOSA Platform

KINESIS exploits five MYOSA strengths: (1) plug-and-play I2C modularity for rapid multi-sensor prototyping; (2) ESP32 dual-core for simultaneous fusion and communication; (3) open-source firmware for deep pipeline customization; (4) built-in Bluetooth for seamless phone AI integration; (5) stackable design for cascading multiple IMUs. The project extends MYOSA from educational use into clinical healthcare IoT, demonstrating its medical sensor prototyping potential.

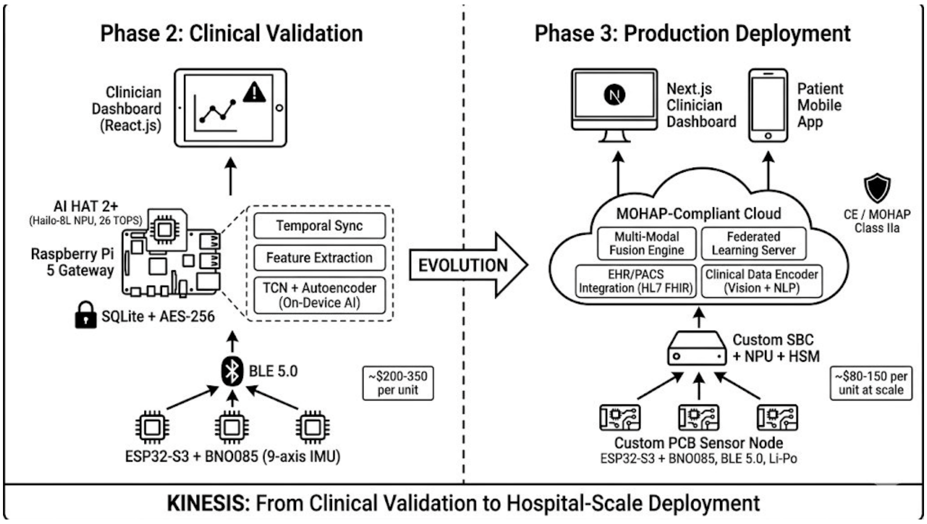

Fig. 2. Evolution: Phase 2 (RPi 5 + AI HAT 2+, clinical validation) and Phase 3 (custom PCB, hospital deployment).

D.9 Demonstration Plan

The live demo (3-5 min) showcases the full pipeline. A team member wears two MYOSA nodes (upper arm, forearm) and performs shoulder abduction, elbow flexion/extension, and sit-to-stand. The audience observes: real-time OLED display of ROM and quality score; phone app with multi-axis visualization and AI classification with confidence; barometric sit-to-stand detection; trend arrows showing session improvement; and a generated PDF clinician report. The demo concludes with the Phase 2/3 roadmap overview.

References

[1] WHO, "Musculoskeletal health," Fact Sheets, 2022. [Online]. Available: who.int/news-room/fact-sheets/detail/musculoskeletal-conditions

[2] A. Cieza et al., "Global estimates of the need for rehabilitation based on the GBD study 2019," The Lancet, vol. 396, no. 10267, pp. 2006-2017, 2021.

[3] S. Safiri et al., "The global macroeconomic burden of musculoskeletal disorders," Rheumatol. Int., 2025.

[4] GBD 2021 Collaborators, "Global burden of musculoskeletal disorders, 1990-2021, and forecast to 2035," Front. Public Health, vol. 13, 2025.

[5] B. Gu et al., "Advancements in wearable sensor technologies for health monitoring: systematic review," JMIR mHealth uHealth, vol. 14, no. 1, e76084, 2026.

[6] M. A. de Cavalieri et al., "Trends in wearable technology for motor rehabilitation," Sensors, vol. 24, 2024.

[7] Fierce Healthcare, "Sword Health buys Kaia Health in $285M deal," Jan. 2026; Healthcare Technology Report, "Top 25 Digital Health Executives," 2025.

[8] SNS Insider, "3D motion capture system market to reach $726.08M by 2032," Sep. 2024.

[9] PMC, "Musculoskeletal digital therapeutics and digital health rehabilitation," 2025.

[10] Healthcare Digital, "The digital MSK market transformation forecast 2026-2035," Dec. 2025.

[11] Nexdigm, "UAE healthcare market report," 2025.